Mevrsic Trust Authority (MTA)

A Decentralised Trust Framework for Embodied Artificial Intelligence

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Part I — Institutional Proposition

1. Executive Summary

Artificial intelligence is moving beyond software environments and into physical bodies.

As AI systems acquire the ability to interact with the physical world through robotics, autonomous machines and embodied platforms, a new category of trust and security infrastructure becomes necessary.

A compromised software agent can expose information, manipulate digital transactions or disrupt services. A compromised embodied AI system may additionally manipulate objects, move through physical environments, interact with people and operate machinery.

Existing digital identity, cybersecurity and robotics safety frameworks address parts of this problem. However, the ecosystem needs a coordinated way to establish and verify trust relationships between physical platforms, AI operators, manufacturers, software providers, administrators and independent security organisations.

The **Mevrsic Trust Authority (MTA)** is proposed as an independent, decentralised trust institution that provides this coordination.

MTA will establish the governance, identity infrastructure, credential frameworks, verification mechanisms and trust registries necessary for secure interoperability between embodied AI systems and their operators.

Its fundamental proposition is:

An embodied AI system must be able to verify whom it trusts, establish what those entities are authorised to do, and retain independent control over its physical safety.

MTA will not own or control participating robots. It will establish a framework through which trust can be independently established, verified, delegated, audited and revoked.

2. Vision

To establish a globally interoperable trust ecosystem for embodied artificial intelligence, enabling physical AI platforms to operate securely across manufacturers, AI providers, operators and institutional boundaries.

3. Mission

MTA's mission is to:

1. Establish interoperable decentralised identity standards for embodied AI.
2. Authenticate the identity and provenance of physical AI platforms.
3. Establish verifiable trust relationships between hardware, firmware, AI providers and operators.
4. Define capability-based authorisation and permission delegation.
5. Provide trusted registries of recognised credential issuers and security auditors.
6. Establish credential revocation, key recovery and trust-update mechanisms.
7. Promote independent security assessment and certification.
8. Enable AI-provider-independent and hardware-independent interoperability.
9. Support responsible deployment of embodied AI through transparent governance and accountability.

4. The Problem MTA Addresses

The embodied AI ecosystem faces several interconnected problems.

4.1 Identity fragmentation

Robotics manufacturers, AI providers and software platforms maintain different identity systems. There is no universal mechanism through which a physical AI platform can establish interoperable trust across these environments.

4.2 Unclear authority

Authentication alone does not establish what an AI is permitted to do. A verified AI provider should not automatically receive unrestricted authority over a physical machine.

4.3 Supply-chain uncertainty

A device may contain hardware, firmware, operating systems, software libraries and AI components from multiple suppliers. Establishing the provenance and integrity of these components is essential.

4.4 Remote compromise

Internet-connected embodied systems introduce risks involving stolen credentials, compromised providers, malicious updates, unauthorised sessions and communication-channel attacks.

4.5 Lack of portable trust

An AI should be able to operate across compatible physical platforms without requiring the physical platform to surrender its identity, security architecture or safety authority.

4.6 Inconsistent assurance

Manufacturers and AI providers may make security claims without a common framework for independent verification.

MTA addresses these problems through a common trust and governance infrastructure.

Part II — Organisational Structure

5. Institutional Model

MTA should operate as an independent, multi-stakeholder institution.

It should not be owned or exclusively governed by a single robotics manufacturer, AI provider or hardware supplier.

Its institutional structure should separate:

- Governance and policy.
- Technical standards.
- Trust registry operations.
- Credential issuance.
- Independent assessment.
- Dispute resolution and oversight.

This separation is essential to avoid conflicts of interest.

6. Proposed Organisational Structure

6.1 Board of Governors

The Board of Governors provides institutional oversight and establishes the organisation's strategic direction.

Responsibilities:

- Approve institutional strategy.
- Approve governance policies.
- Oversee financial and operational integrity.
- Appoint executive leadership.
- Approve major changes to the trust framework.
- Ensure stakeholder representation.
- Maintain institutional independence.

Proposed representation:

- Embodied AI and robotics manufacturers.
- AI developers and providers.
- Cybersecurity organisations.
- Digital identity specialists.
- Independent researchers.
- Civil society and public-interest representatives.
- Safety and standards organisations.

No single stakeholder category should have unilateral control.

6.2 Executive Directorate

Responsible for the day-to-day administration and implementation of MTA's mandate.

Proposed offices:

Office of the Executive Director

- Institutional administration.
- Strategic execution.
- External relationships.
- Financial management.

Office of Technology

- Technical architecture.
- DID and credential interoperability.
- Protocol development.

- Infrastructure management.

Office of Security and Assurance

- Security policies.
- Threat modelling.
- Incident coordination.
- Audit programme oversight.

Office of Governance and Compliance

- Institutional policies.
- Issuer admission requirements.
- Dispute procedures.
- Regulatory coordination.

Office of Ecosystem Development

- Manufacturer partnerships.
- AI provider participation.
- Developer programmes.
- Public adoption and education.

6.3 Technical Standards Council

An independent technical body responsible for developing and maintaining the MTA technical specifications.

Working groups:

1. Decentralised Identity and DID Methods.
2. Verifiable Credentials and Trust Registries.
3. Hardware Identity and Provenance.
4. Firmware and Software Integrity.
5. AI Operator Identity and Delegation.
6. Capability-Based Authorisation.
7. Security and Recovery.
8. Interoperability and Conformance Testing.

6.4 Independent Security and Assurance Council

Responsible for maintaining the independence and quality of security assessments.

Responsibilities:

- Define audit requirements.
- Recognise qualified assessment organisations.
- Establish testing methodologies.
- Review certification disputes.
- Publish security advisories.
- Coordinate vulnerability disclosure.
- Maintain assessment independence.

The council should not directly certify products that its own members have developed or supplied.

6.5 Ethics, Safety and Public Interest Council

Responsible for reviewing the broader consequences of embodied AI deployment.

Areas of responsibility:

- Physical safety.
- Human interaction.
- Privacy and data minimisation.
- Accountability.
- Responsible AI operation.
- Accessibility.
- Public-interest considerations.

This council advises governance and standards bodies without replacing technical safety engineering or regulatory authorities.

Part III — Technical Architecture

7. Core Architecture

MTA is organised around a trust framework with multiple independently governed components.

Component 1 — Decentralised Identity Infrastructure

Purpose: Establish persistent and verifiable identities for participating entities.

Identity classes:

- Physical Mevrsic platforms.
- Other participating robotic platforms.
- AI providers.
- AI operating instances.
- Hardware manufacturers.
- Firmware developers.
- Administrators and owners.
- Credential issuers.
- Security auditors.

Technical foundations:

- W3C Decentralized Identifiers (DIDs).
- DID methods appropriate to the deployment environment.
- Cryptographic key binding.
- Hardware-backed key storage where available.
- Key rotation and recovery.
- DID document lifecycle management.

MTA should support multiple compatible DID methods rather than require a single universal method.

Component 2 — Verifiable Credential Infrastructure

Purpose: Allow entities to make independently verifiable claims about identity, capabilities, provenance and compliance.

Proposed credential classes:

- Manufacturer Credential.
- Hardware Authenticity Credential.
- Device Registration Credential.

- Firmware Provenance Credential.
- Software Integrity Credential.
- AI Provider Credential.
- AI Operator Credential.
- Administrator Credential.
- Security Assessment Credential.
- Compliance Credential.
- Capability Delegation Credential.

Technical foundation: W3C Verifiable Credentials Data Model.

MTA should define credential profiles, issuer requirements, verification rules and lifecycle policies.

Component 3 — Mevrsic Trust Registry

Purpose: Maintain information about recognised issuers, their authority, roles and applicable governance policies.

The registry should contain:

- Recognised credential issuers.
- Issuer public verification information.
- Issuer roles and scopes.
- Approved credential schemas.
- Governance policies.
- Recognition and suspension status.
- Revocation information.
- Security auditor recognition.
- Framework and policy versions.

The trust registry does not need to issue every identity itself.

It determines which issuers and credentials are trusted for particular purposes.

Component 4 — Hardware Identity and Provenance Authority

Purpose: Establish trust in the physical platform and its manufacturing history.

Functions:

- Manufacturer identity verification.
- Hardware authenticity credentials.
- Device registration.
- Component provenance.
- Hardware security capability declarations.
- Hardware lifecycle records.
- Tamper-resistance claims.

A device should be able to establish its identity independently of the AI currently operating it.

Component 5 — Firmware and Software Integrity Authority

Purpose: Establish confidence in the software executing on embodied platforms.

Functions:

- Firmware signing policies.
- Secure boot requirements.
- Software provenance credentials.
- Update integrity verification.
- Version and compatibility attestations.
- Vulnerability and support-status information.
- Recovery and rollback policies.

MTA should recognise qualified software and firmware authorities without becoming the sole firmware-signing organisation.

Component 6 — AI Provider and Operator Authority

Purpose: Establish verifiable identities for AI providers and authorised operating instances.

Functions:

- AI provider registration.
- Provider identity credentials.
- Operator instance identity.
- Deployment identity.
- Provider security declarations.

- AI session identity.
- Operator credential revocation.

An AI provider's identity does not automatically grant its models physical authority.

Component 7 — Capability and Delegation Framework

Purpose: Define what an authenticated AI is allowed to do.

Proposed capability categories:

- Observe.
- Communicate.
- Manipulate.
- Navigate.
- Interact with approved tools.
- Administer permitted configurations.
- Perform maintenance under authorised procedures.

Permissions should be:

- Explicitly granted.
- Limited in scope.
- Time-bound where appropriate.
- Revocable.
- Auditable.
- Restricted by local physical safety policies.

Capability credentials should not be interpreted as permission to bypass a platform's independent safety controller.

Component 8 — Revocation and Recovery Infrastructure

Purpose: Ensure that trust can be withdrawn and restored securely.

Functions:

- Credential revocation.
- Issuer suspension.
- Key compromise handling.

- Device compromise status.
- AI operator disconnection.
- Credential replacement.
- Recovery authority procedures.
- Trust-policy updates.

Revocation must be possible without requiring the compromised entity to cooperate.

Component 9 — Security Assessment and Certification

Purpose: Provide independent assurance that products conform to MTA requirements.

Assessment domains:

- Identity and authentication.
- Hardware security.
- Firmware integrity.
- Communications security.
- AI session security.
- Capability enforcement.
- Physical safety architecture.
- Incident response.
- Recovery procedures.
- Privacy and auditability.

MTA should recognise independent assessors and establish assessment criteria.

Certification should describe precisely what was tested, under which version of the framework, and what limitations remain.

Component 10 — Audit and Accountability Infrastructure

Purpose: Maintain verifiable records of important trust and security events.

Proposed records:

- Identity registration.
- Credential issuance.
- Permission delegation.

- Permission revocation.
- Firmware attestation.
- Security assessment.
- Administrative changes.
- Incident reports.
- Recovery operations.

Audit records should be tamper-evident and privacy-preserving.

Sensitive operational information should not automatically be published to public registries.

Part IV — Authority and Security Model

8. Separation of Trust and Physical Authority

One of MTA's most important principles is that establishing trust must not confer unrestricted physical control.

The proposed model separates four functions:

1. **Identity:** Who is the entity?
2. **Trust:** Which claims and issuers do we recognise?
3. **Authorisation:** What is the entity permitted to do?
4. **Execution:** Is the requested action safe and permissible under the local platform's rules?

The Mevrsic platform retains independent physical authority through its local safety controller.

MTA establishes and verifies trust relationships. It does not directly command motors, actuators or other physical mechanisms.

9. Security Principles

MTA should adopt the following principles:

Zero-trust architecture

No entity is trusted solely because it is inside a network, belongs to a recognised provider or has previously been authenticated.

Least privilege

Every AI operator receives only the permissions necessary for its authorised role.

Hardware-rooted trust

Critical cryptographic identities should use protected hardware facilities where feasible.

Secure lifecycle

Identity, software, firmware and credentials must be managed throughout the platform's lifecycle.

Independent safety enforcement

Physical safety restrictions must remain independent of external AI providers and MTA infrastructure.

Compromise containment

A compromised credential, operator or provider should not automatically compromise the entire platform.

Recovery by design

Every participating platform should have a documented and tested recovery process.

Privacy by design

Identity and trust verification should disclose only the information necessary for the verification purpose.

Part V — Governance and Membership

10. Membership Categories

MTA should offer several participation levels.

Founding Members

Organisations participating in the initial institutional and technical establishment of MTA.

Hardware and Robotics Members

Manufacturers and suppliers of embodied AI platforms and components.

AI Provider Members

Organisations providing AI models, agents and operating services.

Technology Members

Developers of identity infrastructure, cryptographic systems, security software and interoperability tools.

Issuer Members

Organisations authorised to issue defined classes of verifiable credentials.

Security and Assurance Members

Independent cybersecurity assessment and certification organisations.

Research Members

Universities, laboratories and independent research organisations.

Public Interest Members

Organisations contributing to safety, ethics, privacy and public-interest governance.

Membership does not automatically confer credential-issuing authority or certification rights.

11. Issuer Admission and Governance

Credential issuers must be recognised according to defined requirements.

Proposed requirements:

- Verified institutional identity.
- Defined issuing authority.
- Published credential schemas.
- Secure signing infrastructure.
- Key management and recovery policies.
- Audit requirements.
- Revocation mechanisms.
- Conflict-of-interest disclosures.
- Periodic reassessment.

Issuer recognition should be scoped.

An organisation recognised to issue hardware credentials should not automatically be recognised to certify AI providers or security assessments.

12. Institutional Independence

MTA should maintain:

- Transparent governance.
- Independent technical standards.

- Separation of issuer and assessor functions.
- Published certification criteria.
- Documented appeals procedures.
- Conflict-of-interest policies.
- Public change records.
- Periodic independent audits.

MTA should avoid becoming a commercial gatekeeper that controls which AI systems or manufacturers may exist.

Its role is to provide a trustworthy interoperability framework, not to monopolise embodied AI.

Part VI — Operational Model

13. Example: A Mevrsic Platform Joins the Ecosystem

A manufacturer produces a new Mevrsic platform.

1. The manufacturer registers its institutional identity.
2. The hardware receives a persistent device identity.
3. Hardware provenance credentials are issued.
4. Firmware integrity information is registered or attested.
5. The device obtains the relevant trust registry information.
6. An AI provider presents its operator credentials.
7. The device verifies the provider and operator.
8. A capability credential delegates limited permissions.
9. The local safety controller independently validates physical actions.
10. Relevant trust and security events are recorded.

The same physical platform can subsequently accept another authorised AI provider without changing its persistent device identity.

14. Example: Compromised AI Operator

Suppose an authorised AI provider experiences a security incident.

1. The incident is identified.

2. The affected operator identity or credentials are suspended or revoked.
3. Participating platforms receive updated trust information.
4. Existing sessions are terminated according to local security policy.
5. The physical platform enters or maintains its defined safe operating state.
6. Audit records preserve the relevant events.
7. A replacement operator must establish a new authenticated session.
8. New capabilities are delegated only after verification.

The platform's identity and local safety infrastructure remain intact.

Part VII — Implementation Roadmap

15. Phase One: Foundation

Objectives:

- Establish the institutional charter.
- Define the initial governance structure.
- Adopt the core technical principles.
- Select supported DID methods.
- Define initial credential profiles.
- Develop the first trust registry specification.
- Establish the security and assurance requirements.

Deliverable: MTA Framework v1.0.

16. Phase Two: Technical Prototype

Objectives:

- Implement a DID-based device identity.
- Implement AI operator identity.
- Build a prototype trust registry.
- Implement credential issuance and verification.
- Demonstrate capability delegation.

- Demonstrate credential revocation.
- Integrate an independent local safety controller.

Deliverable: MTA Reference Implementation.

17. Phase Three: Pilot Ecosystem

Objectives:

- Partner with selected robotics manufacturers.
- Onboard AI providers.
- Recognise initial credential issuers.
- Conduct independent security assessments.
- Test provider switching and identity continuity.
- Test compromise containment and recovery.

Deliverable: MTA Pilot Programme.

18. Phase Four: Institutional Establishment

Objectives:

- Establish the independent legal entity.
- Formalise membership and governance.
- Introduce issuer accreditation.
- Establish recognised security assessment programmes.
- Publish interoperability specifications.
- Begin broader ecosystem adoption.

Deliverable: Operational MTA.

19. Phase Five: Ecosystem Expansion

Objectives:

- Expand support beyond Mevrsic hardware.
- Establish cross-manufacturer interoperability.
- Develop international partnerships.
- Participate in standards organisations.

- Expand credential and assurance categories.
- Support additional embodied AI environments.

Deliverable: An interoperable trust ecosystem for embodied AI.

Part VIII — Institutional Value Proposition

20. Why Participate in MTA?

For robotics manufacturers

- Establish verifiable hardware identity.
- Demonstrate platform provenance.
- Support interoperability across AI providers.
- Reduce duplicated identity infrastructure.
- Improve customer confidence.

For AI providers

- Establish portable operator identities.
- Demonstrate security and operational credentials.
- Integrate with multiple hardware platforms.
- Delegate capabilities through common mechanisms.
- Reduce dependence on proprietary hardware integrations.

For cybersecurity organisations

- Participate in a defined assurance ecosystem.
- Provide independent assessments.
- Establish recognised issuer credentials.
- Contribute to emerging embodied AI security standards.

For developers

- Build against interoperable identity and trust interfaces.
- Avoid implementing separate trust systems for every manufacturer.

- Use standard credential and delegation models.

For governments and public institutions

- Access verifiable claims about participating platforms.
- Establish procurement and assurance requirements.
- Support responsible deployment.
- Participate in governance without controlling the entire ecosystem.

Part IX — Strategic Positioning

21. What MTA Is—and Is Not

MTA is:

- A decentralised trust framework.
- A governance and standards institution.
- A trust registry operator.
- An ecosystem for recognised credential issuers.
- A framework for identity, provenance, authorisation and assurance.
- An interoperability layer for embodied AI.

MTA is not:

- A central controller of participating robots.
- A universal AI identity provider.
- A replacement for hardware manufacturers.
- A replacement for independent safety engineering.
- A substitute for government regulation.
- A guarantee that a certified system can never be compromised.
- A blockchain project whose primary purpose is token issuance.

MTA should not require a native cryptocurrency or speculative token economy. Its economic sustainability should come from institutional membership, technical services, certification programmes and ecosystem participation.

Part X — Foundational Principles

22. MTA Charter Principles

1. Persistent Identity

Every participating embodied platform should retain a verifiable identity independent of its current AI operator.

2. Interoperability

Trust relationships should function across compatible manufacturers, AI providers and technology environments.

3. Delegated Authority

Authentication must not imply unrestricted authority. Permissions must be explicit and scoped.

4. Independent Physical Safety

No external identity or trust provider should be able to override a platform's independent safety controls.

5. Decentralised Trust

Trust should be distributed among recognised, accountable issuers rather than monopolised by a single identity provider.

6. Verifiable Provenance

Hardware, firmware, software and operator claims should be independently verifiable.

7. Revocability

Trust and permissions must be withdrawable when identities, keys or systems become compromised.

8. Accountability

Important security and governance decisions must be auditable.

9. Privacy

Verification should not require unnecessary disclosure of personal, operational or commercially sensitive information.

10. Institutional Neutrality

MTA should remain independent of individual AI providers and robotics manufacturers.

Closing Proposition

The next generation of artificial intelligence will not be confined to individual software platforms.

AI systems will operate through physical machines, migrate between compatible embodiments, interact with the physical world and participate in increasingly complex environments.

This transition requires a trust infrastructure that is as interoperable as the AI ecosystem itself.

The Mevrsic Trust Authority proposes such an infrastructure.

By combining decentralised identity, verifiable credentials, trusted issuer registries, capability-based authorisation, hardware and firmware provenance, independent security assessment and revocation mechanisms, MTA seeks to establish a common foundation for secure embodied AI.

Its defining principle is simple:

Trust the identity. Verify the claims. Limit the authority. Preserve the body's independent safety.

MTA's objective is not to decide which intelligence should inhabit a physical platform. It is to ensure that whichever intelligence operates that platform does so within a verifiable, accountable and secure framework.

Mevrsic Trust Authority — Establishing trust for the age of embodied intelligence.